

Rice Tungro Disease: From Identification to Disease Control

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Abstract: As the most devastating viral disease of rice in South and Southeast Asia, rice tungro disease remains one of the significant fears to sustainable annual rice productions in the world. Due to the increasing world population and subsequent increase in demand for food, identifying the causal agents and symptoms of disease and the methods of disease management are key to understanding how to reduce the economic damage caused by rice pathogens. In this review, we described the current state of knowledge of rice tungro disease caused by two morphologically and genomically dissimilar viruses: *Rice tungro bacilliform virus* and *Rice tungro spherical virus*. This includes genome structure, transmission, symptoms, diagnostic methods and biological control of the disease.

Key words: *Oryza sativa* • Rice Tungro • Rice tungro bacilliform virus • Rice tungro spherical virus

INTRODUCTION

Asian cultivated rice, *Oryza sativa* L., belongs to the family of Poaceae (Gramineae) and is one of the world's most important staple crops for a third of the human population. As rice is the primary food for most Asian countries and is consumed by 2.9 billion Asians, Asia produces more than 90% of world's rice, with China as the largest producer followed by India, Indonesia, Bangladesh and Vietnam [1]. In major rice-growing countries, the outbreaks of rice disease remain the major threat to sustainable rice production.

Rice tungro disease (RTD) is widely distributed in South and South-east Asian countries and has been recognised as a serious constraint to rice production. This disease was considered as a nutritional disorder of rice in the early 1950's and the catastrophic epidemics significantly devastated the rice production industry. In the Filipino dialect, the word "Tungro" means degenerated growth and the disease has been called "Penyakit merah" in Malaysia, "yellow-orange leaf" in Thailand, "mentek" or "habang" in Indonesia and "accepna pula" in the Philippines. A series of outbreaks were recorded in several rice production countries such

as India, Indonesia, Malaysia, Philippines, China, Thailand and Bangladesh. As one of the most destructive disease, RTD was found to cause a worldwide annual loss in rice production of approximately US \$1.5 billion and 5% to 10% reduction in rice yields in South and Southeast Asia [2]. In Indonesia, this devastating virus affected about 199 000 ha of rice crops between 1968 and 1994 [3].

Rice tungro disease was found to be associated with two distinct viruses: *Rice tungro bacilliform virus* (RTBV) and *Rice tungro spherical virus* (RTSV) [4]. RTBV is a double stranded (ds) DNA genome virus and a member of the *Tungrovirus* genus in the *Caulimoviridae* family with particles sizes of 100-300 nm in length and 30-35 nm in width. On the other hand, RTSV is a single-stranded (ss) RNA virus and a member in the genus of *Waikavirus* (*Sequiviridae*). Virus particles are polyhedral and about 30 nm in diameter [5]. Interestingly, tungro disease can be caused either by a single or mixed infection of RTBV and RTSV, however, rice (varieties TN1 and FK-135) infected with RTBV alone or infected with both RTSV and RTBV produced significant yield reductions (more than 85%) but these yield reductions were not seen when infected with RTSV alone [6]. RTSV acts as a helper virus for transmission of RTBV and

simultaneous infection with these two viruses allows disease development and full symptom expression to occur.

Basic Genome of the RTBV and RTSV: RTSV is a single-stranded, positive-sense RNA virus with genome of approximately 12.5 kb. Its genome encodes polypeptides of more than 390 kDa and has two short open reading frames at the 3' end [5, 7]. A putative leader protein (72kDa), three coat proteins (CP1, CP2 and CP3), a 3C-like protease, a nucleotide polymerase and a polymerase are all contained in the large polypeptide [5, 7]. Conversely, RTBV has a circular double-stranded genome of approximately 8 Kb and this virus is named after its bacilliform particles [5, 8]. As the only member of the genus *Tungrovirus*, the RTBV genome is transcribed asymmetrically, with its coding capacity on a single strand, which contains four open reading frames (ORFs) and two site-specific discontinuities in comparison with other *Badnaviruses* [9]. Genus *Badnavirus* differs to *Tungrovirus* by only having three ORFs and having mealy bugs as transmission vectors. The corresponding proteins from the four ORFs are P1 (24kDa), P2 (12 kDa), P3 (194 kDa) and P4 (46 kDa). These frames encode a CP, protease, movement protein and a reverse transcriptase/RnaseH, as well as proteins of unknown function [5, 8, 10-12]. At present, the genome sequences of RTBV have been found to be clustered into two distinct groups, South East Asian and South Asian groups [13]. Further recent developments in rice tungro virus genomes have focused on understanding the RNA recombination mechanism, the molecular biology of interaction between the two viruses as well as analysis

of significant motif and domains indicating their biological importance [2, 13].

Transmission: Tungro disease was discovered as a leafhopper transmitted virus in 1965 [14]. The tungro disease is transmitted by six leafhopper species in a semi-persistent manner and *Nephotettix virescens* (dist.) is known as the predominant vector [15]. The virus retention period for the green leafhopper (GLH) is two to four days for RTSV and four to five days in the case of RTBV. The GLH readily gains RTSV on source plants infected with RTSV alone, but does not gain RTBV on plants infected only by RTBV [16]. The transmission of RTBV is dependent on the presence of RTSV, thus GLH that previously fed on plants infected by RTSV and subsequently obtained RTSV can then acquire RTBV from a plant infected with RTBV [17-18].

Symptoms and Diagnostic Methods: Generally tungro diseases are not easy to identify in the field due to confusion with disorders induced by abiotic and biotic factors as well as different symptoms that are being expressed depending upon the rice cultivars, the presence of virus particle(s), rice growth stages and growth conditions of the rice plants [19]. The typical symptoms of rice infected with RTBV and RTSV are stunting, yellow or yellow to orange discoloration of infected leaves, reduced tillering, sterile panicles and often irregular-shaped darkbrown specks are visible on the leaves (Figure 1). The young infected leaves may have a mottled appearance and interveinal chlorosis, whilst old leaves will show rust-coloured specks of varying size.



Fig. 1: Rice infected with tungro in Kedah, Malaysia.

RTBV infection alone produces similar but milder tungro symptoms in the infected plants. RTSV alone causes indistinct symptoms such as mild stunting. As the plant ages, the percentage of infection decreases at the time of infection but the latent period of virus infection increases [20-21]. Resistant cultivars infected with either virus exhibit delayed flowering whereas susceptible rice plants do not produce flowers. Visual observations of typical rice tungro disease symptoms is a common practical method for virus detection in the field, although sometimes the disease is misdiagnosed and misidentified as a non-pathogenic disorder, because the symptoms produced are similar to symptoms exhibited following overwatering, nutritional deficiencies or insect damage.

As symptom observation in the field to detect the viruses is not always reliable, several diagnostic methods have been developed for the detection of RTBV and/or RTSV. The most common, specific and relatively reliable method is based on a serological method of detection [22-23]. The four serological methods: enzyme-linked immunosorbent assay (ELISA), simplified ELISA, the latex flocculation test (LF) and the passive hemagglutination test (PHA) have been used for detection of RTBV and RTSV in both rice plants and insect vectors [23]. In the study, ELISA was found to have highest detection sensitivity followed by simplified ELISA, LF and PHA. However, none of the methods used in the study are able to detect RTSV and RTBV antigens in their respective vector insects. It might be that the sensitivity of ELISA is not high enough to detect the rice viruses in insect vectors with low virus concentration.

The rapid and the most sensitive technique that has been developed to detect low levels of extracted RTBV DNA from leaf samples is polymerase chain reaction (PCR) [24]. Takahashi *et al.* [24] demonstrated the effectiveness of PCR for evaluating rice cultivars that were tolerant to RTBV and this technique was found to be 10^3 - 10^4 more sensitive than ELISA. Dasgupta *et al.* [25] later simplified the methods of detection of field infection of RTBV directly from leaf extracts without preparation of DNA. Reverse transcriptase PCR and RFLP analysis have been carried out with viral RNA for differentiation of two RTSV variants [26].

Periasamy *et al.* [27] demonstrated the detection of both viruses in a single multiplex RT-PCR amplification using first strand cDNA as a template. A reverse transcription-loop mediated isothermal amplification (RT-LAMP) assay was established by Le *et al.* [28] for detection of both RTBV and RTSV as well as in viruliferous insect vectors. In a RT-LAMP assay, two

pairs of primers that recognize six regions were designed based on the conserved regions of the nucleotide sequence for each virus. ORF3 and ORF1 regions were used as the target segment for RTBV and RTSV for detection using RT-LAMP, respectively. Recently, SYBR Green 1-based real-time PCR was developed for quantitative determination of RTBV and RTSV in an infected plant [29].

Control

Insecticides: One of the significant protections against tungro disease is the use of insecticides, although the efficiency of insecticides against tungro through vector control is considered to be low. In addition, the use of insecticides is non-specific, killing non-target organisms and causing environmental pollution as well as emergent vector resistance [30-32]. Despite of these disadvantages, use of insecticides is seen to be essential in order to successfully control *N. virescens*, especially after the outbreak of another grass plant hopper, *Nilaparvata lugens* (Stal), following which regular spraying in most of the countries was recommended in huge quantities, with up to six applications per year suggested, regardless of the level of pest infestation [33].

The use of systemic granular insecticides such as Carbofuran is considered to be most effective against tungro disease as they are long lasting and have rapid activity [34-35]. Insecticide applied to the roots of plants provides the most efficient uptake and also much slower degradation rate [36]. Oils and plant extracts have also been evaluated in order to avoid the toxic effects of compounds including neem (*Azadirachta indica*) and custard apple (*Annona squamosa*). Neem is reported to have antifeedant and insecticidal properties and works well against tungro disease if applied directly into the soil [37]. Natural products are not currently being used in the fight against tungro disease due to cost effectiveness and slow pest killing effect.

Conventional Resistance Breeding: Besides suggesting the use of insecticide to control the vector, the main strategy for control of tungro disease is to grow tungro resistant cultivars [38-39]. More than 80,000 accessions of rice are maintained at the International Rice Research Institute (IRRI). At least a third of these have been tested since 1963 for resistance to tungro using various methods [40-41]. As vector resistance was plentiful and easy to identify in the germplasm, a lot of the IR varieties released after 1969, with exception of IR22, were evaluated as leafhopper resistant rather than resistant to the tungro

viruses. However, this type of resistance was not sustainable due to high disease pressure and very often varieties with vector resistance were defeated by tungro as the vector population become adapted after their release [42-43]. As a result of this, many genetic studies have been focused on understanding the inheritance virus resistance [44-47].

Genetically Engineered Resistance against RTSV, RTBV and Green Leaf Hopper (GLH): Several efforts have demonstrated acquired resistance against RTBV, RTSV and GLH using transgenic approaches [2, 48]. Interestingly, the discovery of two rice transcription factors, RF2a and RF2b, that have been shown to play a role in virus replication and symptom development, implies that this could provide a new source of candidate genes for engineered resistance [2, 49]. No drastic phenotypic changes were observed in production of overexpressed RF2a and RF2b transcription factors in transgenic lines [50-52]. Surprisingly, if transgenic lines that overexpressed one of these two transcription factors were inoculated with RTBV through *Agrobacterium*-mediated infection, or by both RTBV and RTSV through GLH transmission, RTBV titers were considerably reduced and substantial resistance to RTD was apparent in transgenic lines [49] and this could be a promising result towards a world without rice tungro disease. This result may suggest more research is needed to elucidate the interaction components between the host and pathogens that can be used to engineer disease resistance [53].

CONCLUSIONS

Rice tungro disease has the potential to cause massive losses in rice production and the lack of any completely effective method to control the disease makes tungro a big threat to world food security. In previous decades, our understanding of rice tungro disease has led researchers to focus on more targeted strategies to overcome this disease and this has been possible with use of multidisciplinary research that may point towards new strategies for their management. Enhanced understanding of the identification, genome type, transmission and biological control of these viruses makes tungro disease very significant in terms of plant virology, molecular biology and entomology, with the focus on achieving the ultimate goal of improved management strategies for control of rice tungro disease in order to reduce the economic damage to global rice production.

ACKNOWLEDGMENTS

This review was funded by Ministry of Agriculture (MOA), Malaysia. The authors expressed their thanks to Kathryn Ford for their editorial comments on the paper.

Conflicts of Interest: The authors declare no conflict of interest.

REFERENCES

1. Brar, D.S. and K., Singh, 2011. *Oryza*. In Wild crop relatives: genomic breeding and resources: cereals; Kole, C., Eds; Springer-Verlag, Berlin, Heidelberg, pp: 321-366.
2. Dai, S. and R.N. Beachy, 2009. Genetic engineering of rice to resist rice tungro disease. *In Vitro Cellular Dev. Biol.-Plant*, 45: 517-524.
3. Hassanuddin, A., Koesnang and D. Baco, 1997. Rice tungro virus in Indonesia. Present status and current management strategy. In *Epidemiology and management of rice tungro disease*, Chancellor, T.C.B., Thresh, J.M., Eds.; National Resources Institute: Chatham, U.K., pp: 94-102.
4. Hibino, H., N. Saleh and M. Roechan, 1979. Transmission of two kinds of rice tungro-associated viruses by insect vectors. *Phytopathology*, 69: 1266-1268.
5. Hull, R., 1996. Molecular biology of rice tungro viruses. *Annu. Rev. Phytopathol*, 34: 275-297.
6. Hassanuddin, A. and H. Hibino, 1989. Grain yield reduction, growth retardation and virus concentration in rice plants infected with tungro-associated viruses. *Trop. Agric. Res. Ser.*, 22: 56-73.
7. Shen, P., M. Kaniewska, C. Smith and R.N. Beachy, 1993. Nucleotide sequence and genomic organisation of rice tungro spherical virus. *Virology*, 193: 621-630.
8. Qu, R., M. Bhattacharya, G.S. Laco, A. De Kochko, B.L. SubbaRao, M.B. Kaniewska, J.S. Elmer, D.E. Rochester, C.E. Smith and R.N. Beachy, 1991. Characterization of the genome of rice tungro bacilliform virus: Comparison with Commelina yellow mottle virus and caulimoviruses. *Virology*, 185: 354-364.
9. Bao, Y. and R. Hull, 1993. Mapping the 5'-terminus of rice tungro bacilliform viral genomic RNA. *Virology*, 197(1): 445-448.
10. Laco, G.S. and R.N. Beachy, 1994. Rice tungro bacilliform virus encodes reverse transcriptase, DNA polymerase and ribonuclease H activities. *Proc. Natl. Acad. Set U.S.A.*, 91: 2654-2658.

11. Laco G.S., S.B. Kent and R.N. Beachy, 1995. Analysis of the proteolytic processing and activation of the rice tungro bacilliform virus reverse transcriptase. *Virology*, 208: 207-214.
12. Marmey, P., B. Bothner, E. Jacquot, A. de Kochko, C.A. Ong, P. Yot, G. Siuzdak, R.N. Beachy and C.M. Fauquet, 1999. Rice tungro bacilliform virus open reading frame 3 encodes a single 37-kDa coat protein. *Virology*, 253: 319-326.
13. Banerjee, A., S. Roy and J. Tarafdar, 2011. Phylogenetic analysis of Rice tungro bacilliform virus ORFs revealed strong correlation between evolution and geographical distribution. *Virus Genes*, 43: 398-408.
14. Rivera, C.T. and S.H. Ou, 1965. Leafhopper transmission of 'tungro' disease of rice. *Plant Dis. Rep.*, 49: 127-131.
15. Ling, K.C., 1972. *Rice Virus Diseases*, International Rice Research Institute: Los Banos, Philippines, pp: 93-105.
16. Hibino, H., M. Roechan and S. Sudarisman, 1978. Association of two types of virus particles with penyakit habang (tungro disease) of rice in Indonesia. *Phytopathology*, 68: 1412-1416.
17. Hibino, H. and P.Q. Cabauatan, 1987. Infectivity neutralization of rice tungro associated viruses acquired by vector leafhoppers. *Phytopathology*, 77: 473-476.
18. Cabauatan, P.Q. and H. Hibino, 1985. Transmission of Rice tungro bacilliform and spherical viruses by *Nephotettix virescens* distant. *Philippine Phytopathol.*, 21: 103-109.
19. Azzam, O. and T.C. Chancellor, 2002. The biology, epidemiology and management of rice tungro disease in Asia. *Plant Dis.*, 86: 88-100.
20. Ling, K.C. and M.K. Palomar, 1966. Studies on rice plants infected with the virus at different ages. *Philipp. Agric.*, 50: 165-177.
21. Astika, N.S., N. Suwela, G.N. Astika and Y. Suzuki, 1992. Dependence of incubation period and symptoms of rice tungro disease (RTD) on infected stage in rice fields. *Int. Rice Res. NewsL.*, 17(3): 19-20.
22. Bajet, N.B., V.M. Aguiro, R.D. Daquioag, G.B. Jonson, R.C. Cabunagan, E.M. Mesina and H. Hibino, 1986. Occurrence and spread of rice tungro spherical virus in the Philippines. *Plant Dis.*, 70: 971-973.
23. Takahashi, Y., T. Omura, K. Shohara and T. Tsuchizaki, 1991. Comparison of four serological methods for practical detection of ten viruses of rice in plants and insects. *Plant Dis.*, 75: 458-461.
24. Takahashi, Y., E.R. Tiongeo, P.Q. Cabauatan, H. Koganezawa, H. Hibino and T. Omura, 1993. Detection of rice tungro bacilliform virus by polymerase chain reaction for assessing mild infection of plants and viruliferous leafhoppers. *Phytopathology*, 83: 655-659.
25. Dasgupta, I., B.K. Das, P.S. Nath, S. Mukhopadhyay, F.R. Niazi and A. Varma, 1996. Detection of Rice tungro bacilliform virus in field and glasshouse samples from India using the polymerase chain reaction. *J. Virol. Methods*, 58: 53-58.
26. Yambao, M., P. Cabauatan and O. Azzam, 1998. Differentiation of rice tungro spherical virus variants by RTPCR and RFLP. *International Rice Research Notes*, 23(2): 22-24.
27. Periasamy, M., F.R. Niazi and V.G. Malathi, 2006. Multiplex RT-PCR, a novel technique for the simultaneous detection of the DNA and RNA viruses causing rice tungro disease. *J. Virol. Methods*, 134: 230-236.
28. Le, D.T., O. Netsua, T. Uehara-Ichikia, T. Shimizua, I.R. Choic, T. Omuraa and T. Sasayaa, 2010. Molecular detection of nine rice viruses by a reverse-transcription loop mediated isothermal amplification assay. *J. Virol. Methods*, 170: 90-93.
29. Sharma, S. and I. Dasgupta, 2012. Development of SYBR Green I based real-time PCR assays for quantitative detection of Rice tungro bacilliform virus and Rice tungro spherical virus. *J Virol Methods*, 181: 86-92.
30. Akhtar, M.W., D. Sengupt and A. Chowdhury, 2009. Impact of pesticides use in agriculture: their benefits and hazards. *Interdisc Toxicol.*, 2(1): 1-12.
31. Gilden, R.C., K. Huffling and B. Sattler, 2010. Pesticides and Health Risks. *JOGNN*, 39: 103-110.
32. Satapathy, M.K., 1998. Chemical control of insect and nematode vectors of plant viruses. In *Plant Virus Disease Control*, Hadidi, A., R.K. Khetarpal and Koganezawa, H. Eds.; APS Press: St. Paul, MN, USA, pp: 188-195.
33. Teng, P.S., 1994. Integrated pest management in rice. *Exp. Agric.*, 30: 115-137.
34. Satapathy, M.K. and A. Anjaneyulu, 1984. Use of cypermethrin, a synthetic pyrethroid, in the control of rice tungro virus disease and its vector. *Trop. Pest Manag.*, 30: 170-178.
35. Satapathy, M.K. and A. Anjaneyulu, 1990. Carbofuran as root zone and broadcast application for tungro control: a comparative study. *International Journal of Tropical Plant Diseases*, 8(1): 65-77.

36. Satapathy, M.K. and A. Anjaneyulu, 1989. Experimental epidemics of tungro and its vectors in nursery beds under different pesticide treatments. *International Journal of Tropical Plant Diseases*, 7: 137-150.
37. Saxena, R.C., Z.R. Khan and N.B. Bajet, 1985. Neem seed derivatives for preventing rice tungro virus transmission by the green leafhopper, *Nephotettix virescens* (Distant). *Philipp. Phytopathol.*, 21: 88-102.
38. Imbe, T., H. Hashim, M. Iwasaki and T. Omura, 1993. Resistance in some Japonica rice cultivars to rice tungro spherical virus. *Japan J Breed*, 43: 549-556.
39. Habibuddin, H., K. Hadzim, O. Othman and Y. Azlan, 2000. Y 1286 is a Balimau Putih-derived rice line resistant to rice tungro bacilliform and spherical viruses. *J. Trop. Agric. and Fd. Sc.*, 28(1): 13-22.
40. Hibino, H., R.D. Daquioag, E.M. Mesina and V.M. Aguiro, 1990. Resistances in rice to tungro-associated viruses. *Plant Dis.*, 74: 923-926.
41. Koganezawa, H. and R. Cabunagan, 1997. Resistance to rice tungro virus disease. In *Epidemiology and Management of Rice Tungro Disease*, Chancellor T.C.B. and J.M. Thresh, Eds.; Natural Resources Institute: Chatham, UK, pp: 54-59.
42. Dahal, G., H. Hibino, R.C. Cabunagan, E.R. Tiongco, Z.M. Flores and V.M. Aguiro, 1990. Changes in cultivar reaction to tungro due to changes in "virulence" of the leafhopper vector. *Phytopathology*, 80: 659-665.
43. Manwan, I., S. Sama and S.A. Rizvi, 1985. Use of varietal rotation in the management of tungro disease in Indonesia. *Indones. Agric. Res. Development J.*, 7: 43-48.
44. Nemoto, H. and H. Habibuddin, 1998. Mechanism and inheritance of resistance to rice tungro disease in rice varieties Basmati 370 and IR 50. *Japanese Journal of Tropical Agriculture*, 42(2): 111-118.
45. Azzam, O., T. Imbe, R. Ikeda, N. Nath and E. Coloquio, 2001. Inheritance of resistance to rice tungro spherical virus in a near-isogenic line derived from UtriMerah and in rice cultivar TKM6. *Euphytica*, 122: 91-97.
46. Sama, S., A. Hassanudin, I. Manwan, R.C. Cabunagan and H. Hibino, 1991. Integrated rice tungro disease management in South Sulawesi, Indonesia. *Crop Prot.*, 10: 34-40.
47. Holt, J. and T.C.B. Chancellor, 1997. A model of plant virus disease epidemics in asynchronously-planted cropping systems. *Plant Pathol.*, 46: 490-501.
48. Verma, V., S. Sharma, S.V. Devi, S. Rajasubramaniam and I. Dasgupta, 2012. Delay in virus accumulation and low virus transmission from transgenic rice plants expressing Rice tungro spherical virus RNA. *Virus Genes*, 45(2): 350-359.
49. Dai, S., X. Wei, A.A. Alfonso, L. Pei, U.G. Duque, Z. Zhang, G.M. Babb and R.N. Beachy, 2008. Transgenic rice plants that overexpress transcription factors RF2a and RF2b are tolerant to rice tungro virus replication and disease. *Proc. Natl Acad. Sci. U.S.A.*, 105: 21012-21016.
50. Yin, Y., L. Chen and R. Beachy, 1997a. Promoter elements required for phloem specific gene expression from the RTBV promoter in rice. *Plant J.*, 12: 1179-1188.
51. Yin, Y., Q. Zhu, S. Dai, C. Lamb and R.N. Beachy, 1997b. RF2a, a bZIP transcriptional activator of the phloem-specific rice tungro bacilliform virus promoter, functions in vascular development. *EMBO J.*, 16: 5247-5259.
52. Dai, S., Z. Zhang, S. Chen and R.N. Beachy, 2004. RF2b, a rice bZIP transcription activator, interacts with RF2a and is involved in symptom development of rice tungro disease. *Proc. Natl. Acad. Sci. U.S.A.*, 101: 687-692.
53. Pooggin, M.M., R. Rajeswaran, M.V. Schepetilnikov and L.A. Ryabova, 2012. Short ORF-Dependent Ribosome Shunting Operates in an RNA Picorna-Like Virus and a DNA Pararetrovirus that Cause Rice Tungro Disease. *PLoS Pathog.*, 8(3): e1002568. doi:10.1371/journal.ppat.1002568.